

After we have filled in the justification for these steps, we will have shown that $\sqrt{3}$ is irrational by ~~the~~ adapting the proof that $\sqrt{2}$ is irrational.

Note that this proof can be extended to show that $\sqrt{n}$ is irrational whenever n is a positive integer that is not a perfect square. We leave the details of this to Chapter 4 $\square$

1.8.6 Looking for Counterexamples

Example 17: In Example 15 in Section 1.7 we showed that the statement "Every positive integer is the sum of two squares of integers" is false by finding a counterexample. That is, there are positive integers that cannot be written as the sum of the squares of two integers.

Is the statement "Every positive integer is the sum of the squares of three integers" true or false?

Solution: We will show that 7 cannot be written as the sum of the squares of three integers.

Partitions of 7 as the sum of 3 integers.

- | | | | |
|----|---|---|---|
| 1. | 6 | 1 | 0 |
| 2. | 5 | 2 | 0 |
| 3. | 4 | 3 | 0 |
| 4. | 5 | 1 | 1 |
| 5. | 4 | 2 | 1 |
| | 6 | 3 | 3 |
| | 7 | 3 | 2 |

We can easily see that there does not exist a partition where all three integers are square numbers.

$\therefore$ The statement is False $\square$

Note: It turns out the conjecture "Every positive integer is the sum of the squares of four integers" is true. For a proof, see Elementary Number Theory and its Applications by K.H. Rosen